Telecom Customer Churn Analysis

Executive Summary

Situation

The company is losing roughly one in four customers to churn. This analysis examines customer contract type, tenure, add-on services, and pricing to identify what is driving churn and where retention efforts would have the greatest impact.

Key Findings

- Contract length is the strongest predictor of churn. Month-to-month customers churn at 42.71%, compared to 11.27% for one-year contracts and just 2.83% for two-year contracts - a roughly 15x gap between the least and most committed customer segments.
- Newer customers are far more likely to leave. Customers who churned had an average tenure of just 17.9 months, less than half the 37.5-month average tenure of customers who stayed, showing that churn risk is concentrated early in the customer lifecycle.
- Lack of technical support is strongly linked to churn. Customers without tech support churn at 41.6%, nearly three times the 15.1% churn rate among customers who have it - a stronger effect than either streaming TV or streaming movies showed. Pricing plays a smaller, secondary role: churned customers pay $74.44/month on average versus $61.27/month for retained customers, a real but modest gap compared to the contract and support effects above.

Recommendation

- Prioritize converting month-to-month customers to annual contracts through targeted incentives (e.g., a discount for switching to a one- or two-year plan), since contract length shows the largest single effect on churn.
- Focus retention outreach on customers within their first 12–18 months of tenure, where churn risk is highest, rather than spreading retention budget evenly across the full customer base.
- Bundle free or discounted tech support during onboarding for new customers, given its strong association with lower churn relative to other add-on services.

Business Impact Estimate

The dataset contains 1,655 churned month-to-month customers, who pay an average of $74.44/month. If contract-conversion incentives shifted just 10% of these customers (approximately 166 customers) from churning to retained, the company would retain roughly $12,357 in monthly recurring revenue - approximately $148,000 annualized - before accounting for the cost of the incentives themselves. Since month-to-month customers make up the largest churned segment by far, even a modest conversion rate has a meaningful revenue impact.

Methodology: Data cleaned in Python (Pandas, via Google Colab) - 11 rows were dropped after converting TotalCharges to numeric and removing resulting missing values. All figures in this summary are sourced directly from the SQL queries and results documented in query_results.md, run in DB Browser for SQLite and visualized in Power BI. Dataset: IBM Telco Customer Churn (Kaggle).